Title of the Article

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Abstract

Write a concise summary (150–250 words): context, problem, proposed method, main results, and significance.

Keywords: keyword one; keyword two; keyword three; keyword four

1 Introduction

Context and motivation. Related work can be cited like this [**example2024**]. State the contributions clearly:

- first contribution;
- second contribution;
- third contribution.

The rest of the paper is organized as follows: **??** presents the method, **??** the results, and **??** concludes.

2 Geometry and signal model

Every channel used in this paper is, to first order, a combination of two fixed functions of one normalised coordinate. this section sets the geometry, derives that coordinate and lists what the platform actually measures.

2.1 Survey Geometry

The vehicle moves at speed $v = 1.0\,\mathrm{m\,s^{-1}}$ along straight legs that cross a straight cable at an incidence θ (Figure 1a). Two tri-axial fluxgates are mounted at the water line on the vehicle's lateral axis, UNO on the port side and DUO on the starboard side, separated by the baseline $b = 1\,\mathrm{m}$. For each sensor we use a local CPA frame: the origin is the point of its track directly

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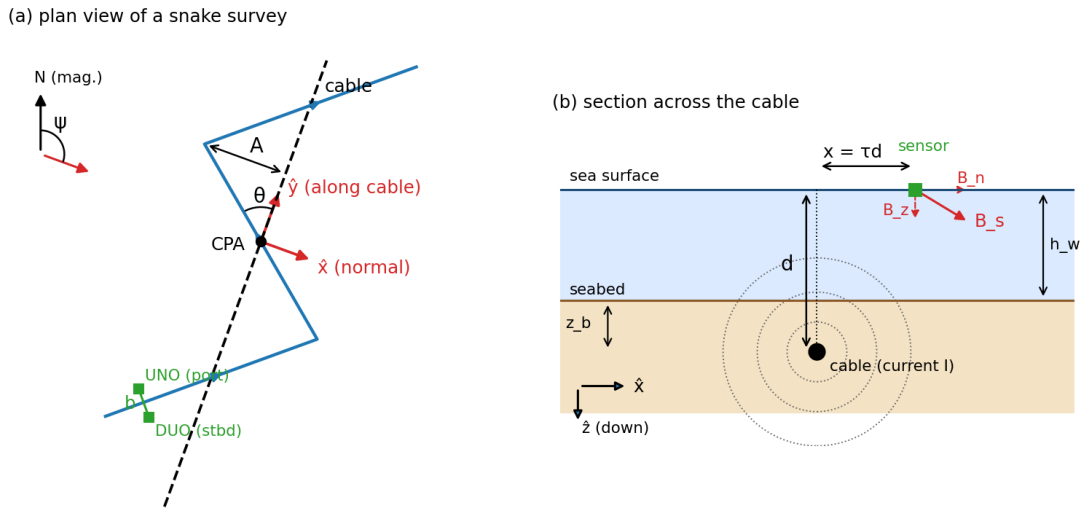


Figure 1: Survey geometry. (a) Plan view of a snake survey: cable (dashed), legs crossing it at incidence θ with amplitude A, CPA frame ($\hat{\boldsymbol{x}}$ normal, $\hat{\boldsymbol{y}}$ along the cable), the two sensors UNO and DUO separated by the baseline b, and the angle ψ from magnetic north to the cable normal. (b) Section across the cable: the sensor at the surface, the cable at standoff d below it (burial depth z_b below a seabed at water depth h_w), the normal coordinate $x = \tau d$, and the azimuthal cable field $\boldsymbol{B}_s$ with its normal and vertical components B_n and B_z.

above the cable, $\hat{\boldsymbol{x}}$ is the horizontal normal to the cable, $\hat{\boldsymbol{y}}$ points along the cable and $\hat{\boldsymbol{z}}$ points down (Figure 1b). The cable then lies at $x = 0$, $z = d$, where d is the standoff.

A sensor that crosses the cable at time t_c has horizontal distance $x = v\,(t - t_c)\sin\theta$ from the cable at time t. Every field quantity below depends on x only through the ratio x/d, so we use the normalised coordinate

$$\tau = \frac{x}{d} = \frac{v\,(t - t_c)\sin\theta}{d} = \frac{t - t_c}{T_s}, \qquad T_s = \frac{d}{v\sin\theta}. \tag{1}$$

τ counts CPA distances, and T_s is the time the sensor takes to move one CPA distance across the cable; at $\theta = 50°$ and $v = 1\,\mathrm{m\,s^{-1}}$ it is 2.6 s for $d = 2\,\mathrm{m}$ and 5.2 s for $d = 4\,\mathrm{m}$. T_s is the time scale of every matched template, and we call it the template scale. A leg that starts and ends at the lateral distance A from the cable covers $|x| \le A$, i.e. $|\tau| \le K$ with

$$K = \frac{L}{d}, \qquad L = A \text{ for a snake leg.} \tag{2}$$

L is the half-length of the pass measured across the cable. With the snake amplitude $A = 6\,\mathrm{m}$ used in closed loop and $d = 2\,\mathrm{m}$ to $4\,\mathrm{m}$, $K = 3$–1.5: a snake leg sees only a few CPA distances on each side of the crossing.

2.2 Cable Field and Cable Orthonormal Basis Functions

We follow the steps of [1]. The cable is modeled as an infinitely long straight conductor carrying a DC current I, with the return spread through the sea far from the sensor. Its field at the

sensor is azimuthal about the cable (Appendix A.1):

$$\mathbf{B}_s(\tau) = K_p \left(\frac{1}{1+\tau^2},\ 0,\ \frac{\tau}{1+\tau^2} \right)^T = (B_n, B_y, B_z)^T, \qquad K_p = \frac{\mu_0 I}{2\pi d}. \tag{3}$$

The components are given in the CPA frame: B_n is along the normal $\hat{x}$, $B_y = 0$ is along the cable, and B_z is downwards. K_p is the field magnitude at CPA, d is the standoff distance and $\tau = x/d$ is the normalized along-track position.

The infinite-line model holds when the cable is straight over several standoffs on each side of the crossing. The simulations instead use the exact field of a polyline made of 2 m segments.

Inserting (3) into the first-order term of the total-field anomaly gives a sum of two fixed functions of τ. Normalised to unit energy, they are the cable OBF of [1]:

$$\Phi_1(\tau) = \sqrt{\frac{2}{\pi}} \frac{1}{1+\tau^2}, \qquad \Phi_2(\tau) = \sqrt{\frac{2}{\pi}} \frac{\tau}{1+\tau^2}. \tag{4}$$

Φ_1 is even and Φ_2 is odd; both have unit energy over the real line, and they are orthogonal(Appendix A.3). The total-field anomaly is then

$$S(\tau) = A_1 \Phi_1(\tau) + A_2 \Phi_2(\tau), \qquad A_1 = \sqrt{\frac{\pi}{2}}\, K_p \cos\delta \cos\psi, \qquad A_2 = \sqrt{\frac{\pi}{2}}\, K_p \sin\delta. \tag{5}$$

The factor $\sqrt{\pi/2}$ undoes the normalisation of (4). A_1 and A_2 are the coordinates of S in the plane spanned by Φ_1 and Φ_2; the matched filter of Section **??** estimates them. The expansion of the norm that justifies keeping only the first-order term, and the resulting phase, are given in Appendix A.2.

Any channel that is a fixed linear combination of the cable field can be written as one vector of $\mathrm{span}\{\Phi_1, \Phi_2\}$:

$$s(\tau) = a \left(\cos\beta\, \Phi_1(\tau) + \sin\beta\, \Phi_2(\tau) \right), \qquad a \geq 0, \quad \beta \in (-180^\circ, 180^\circ]. \tag{6}$$

The amplitude a sets the signal level and the phase β sets the shape: $\beta = 0$ is the even bell Φ_1, $\beta = 90^\circ$ is the odd profile Φ_2, and intermediate phases are asymmetric (Figure 2c).

Hilbert relation. The observation underlying this paper is that Φ_2 is the Hilbert transform of Φ_1. The pair is therefore the real and imaginary parts of a single analytic template:

$$\Phi_2 = \mathcal{H}\{\Phi_1\}, \qquad \Psi(\tau) = \Phi_1(\tau) + i\,\Phi_2(\tau) = \sqrt{\frac{2}{\pi}} \frac{1}{1-i\tau}, \tag{7}$$

and the amplitude and phase of (6) are the modulus and argument of one complex coefficient:

$$s(\tau) = \Re\left\{ a\, e^{-i\beta}\, \Psi(\tau) \right\}. \tag{8}$$

This is the line-source instance of the classical property that the horizontal and vertical components of a 2D potential field form a Hilbert pair. Every channel the platform can form is thus one point $a\, e^{i\beta}$ of the complex plane, and the whole paper is an analysis of what each processing step does to that point.

Two properties of Ψ are used (Appendix A.4). Its spectrum is one-sided, so Φ_1 and Φ_2 have the same autocorrelation; the matched filter of Section **??** is then a quadrature receiver, whose output energy is the squared modulus of a single complex correlation and is therefore independent of β when the record is unbounded. And Ψ continues analytically off the real axis with a single pole at $\tau = -i$, which is what makes the coherence of a window with the span equal to one at the CPA and nowhere else (Appendix **??**). What truncation destroys is the first property, and the displacement it produces is read off the second harmonic $\sin 2\beta$ of the phase.

In the implementation, however, Φ_1 and Φ_2 are used as two real templates and no estimator manipulates complex quantities; the analytic form is a bookkeeping device for the amplitude and the phase.

Gradiometer channels. The gradiometer is the difference DUO − UNO. The lateral axis makes the angle θ with the cable normal, so the two sensors sample the same channel at normal coordinates that differ by $b\cos\theta$, i.e. by $h = b\cos\theta/d$ in τ. To first order in that difference (a central difference),

$$\begin{aligned} \Delta s(\tau) &\simeq \frac{b\cos\theta}{d}\,\frac{\mathrm{d}s}{\mathrm{d}\tau} = \frac{b\cos\theta}{\sqrt{2}\,d}\,a\,(\cos\beta\,\Gamma_1(\tau) + \sin\beta\,\Gamma_2(\tau))\,, \\ \Gamma_1 = \sqrt{2}\,\Phi_1' &= -\frac{4}{\sqrt{\pi}}\frac{\tau}{(1+\tau^2)^2}, \qquad \Gamma_2 = \sqrt{2}\,\Phi_2' = \frac{2}{\sqrt{\pi}}\frac{1-\tau^2}{(1+\tau^2)^2}. \end{aligned} \tag{9}$$

The factor $\sqrt{2}$ makes Γ_1 and Γ_2 unit-energy: with the integrals of Appendix A.3 and $(1-\tau^2)^2 = (1+\tau^2)^2 - 4\tau^2$,

$$\int \Phi_1'^2 = \frac{2}{\pi}\int \frac{4\tau^2}{(1+\tau^2)^4} = \frac{1}{2}, \qquad \int \Phi_2'^2 = \frac{2}{\pi}\left(\int \frac{\mathrm{d}\tau}{(1+\tau^2)^2} - 4\int \frac{\tau^2}{(1+\tau^2)^4}\right) = \frac{1}{2}.$$

Φ_1' is odd and Φ_2' is even, so Γ_1 and Γ_2 are orthogonal by parity (Figure 2b): the vertical gradiometer ΔB_z follows the even profile Γ_2 and the horizontal gradiometer follows the odd profile Γ_1. Since $B_y = 0$ everywhere, the horizontal cable field points along $\hat{x}$ at both sensors, and so does their difference: the horizontal gradiometer vector lies along the cable normal. The first-order model (9) follows from $s(\tau + h/2) - s(\tau - h/2) = h\,s'(\tau) + (h^3/24)\,s'''(\tau) + O(h^5)$, so its relative error is of order $h^2/24$ times the ratio of the third to the first derivative, with $h = 0.21$ at $d = 3$ m and $\theta = 50°$.

The gradiometer amplitude is reduced by $b|\cos\theta|/(\sqrt{2}\,d)$, 0.15 at $d = 3$ m and $\theta = 50°$, and it vanishes at perpendicular incidence. Two departures from (9) are kept in mind. Roll tilts the baseline, giving it the vertical component $b\sin(\text{roll})$; this mixes in the vertical derivative, an impurity of relative size about $\tan(\text{roll})/|\cos\theta|$, 8% at 3° of roll and 50° of incidence, oscillating at the wave period. Differences between the residual calibrations of the two sensors leave a differential term that also varies with attitude.

The channels of this paper have the following amplitudes and phases (with $K' = \sqrt{\pi/2}\,K_p$):

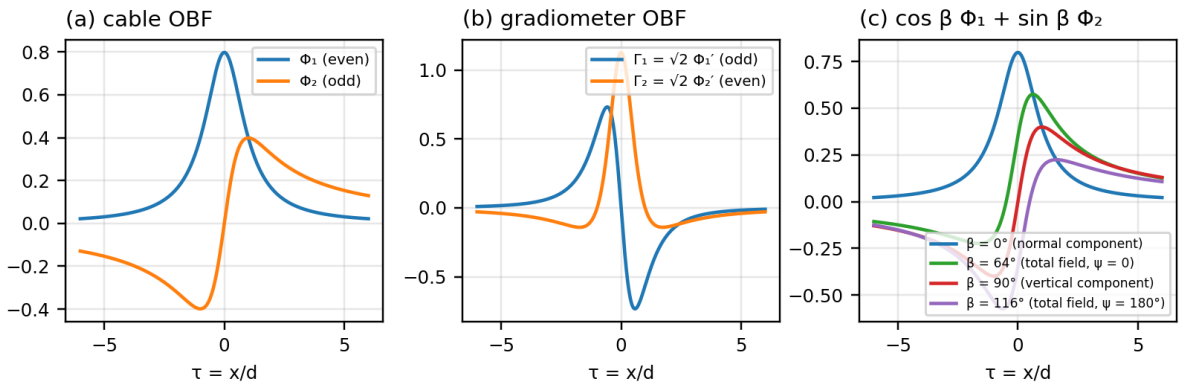


Figure 2: (a) Cable OBF: the even Φ_1 and the odd Φ_2 of (4). (b) Gradiometer OBF: the odd Γ_1 and the even Γ_2 of (9). (c) Channels of (6) for $\beta = 0°$ (normal component), 90° (vertical component), and the limits 64° and 116° of the total-field phase at Brest.

Table 1: Amplitude and phase of each channel in the basis $\{\Phi_1, \Phi_2\}$.

Channel	Amplitude a	Phase β
Normal component B_n	K'	0
Vertical component B_z	K'	90° (−90° for the opposite current)
North and East components, $B_n \cos\psi$ and $B_n \sin\psi$	$K'\lvert\cos\psi\rvert$, $K'\lvert\sin\psi\rvert$	0 or 180°
Total field S	$K'\sqrt{\cos^2\delta\,\cos^2\psi + \sin^2\delta}$	$\tan\beta = \tan\delta/\cos\psi$
Gradiometer, horizontal normal ΔB_n	$\dfrac{K'\,b\,\lvert\cos\theta\rvert}{\sqrt{2}\,d}$	0 on the gradiometer basis (odd profile Γ_1)
Gradiometer, vertical ΔB_z	$\dfrac{K'\,b\,\lvert\cos\theta\rvert}{\sqrt{2}\,d}$	90° on the gradiometer basis (even profile Γ_2)

The vector components have phases fixed by physics. The total-field phase follows from (5), $\tan\beta = A_2/A_1 = \tan\delta/\cos\psi$ (Appendix A.2). Since $\lvert\cos\psi\rvert \le 1$, we have $\lvert\tan\beta\rvert \ge \tan\delta$, so β lies between δ and $180° - \delta$: between 63.7° and 116.3° at Brest, for every cable bearing. The total field is therefore always closer to the odd profile than to the even one. At $\psi = 45°$, for example, $\beta = 70.7°$.

References

[1] Khaled Hleis, Isabelle Quidu, Michel Legris, and Alain Bertholom. “Derivation of New Orthonormal Basis Functions for Magnetic Detection of Submarine Telecommunications Cables”. In: *IET Radar, Sonar & Navigation* 20 (2026), e70225. DOI: 10.1049/rsn2.70225.

A Field of the Cable and the Channel Phases

A.1 Field of an Infinite Line Current

For an infinitely long straight conductor, the Biot–Savart law gives a field of magnitude $\mu_0 I/(2\pi r)$ at the distance r, directed along $\hat{I} \times \mathbf{r}/r$, where $\hat{I}$ is the unit vector of the current and $\mathbf{r}$ the vector from the conductor to the point, perpendicular to it. In the CPA frame the cable runs along $\hat{y}$ at $x = 0$, $z = d$, and the sensor is at $(x, \cdot, 0)$, so $\mathbf{r} = (x, 0, -d)$ and $r^2 = x^2 + d^2$. For a current along $-\hat{y}$, using $\hat{y} \times \hat{x} = -\hat{z}$ and $\hat{y} \times \hat{z} = \hat{x}$,

$$(-\hat{y}) \times (x\,\hat{x} - d\,\hat{z}) = d\,\hat{x} + x\,\hat{z}, \qquad \mathbf{B}_s = \frac{\mu_0 I}{2\pi(x^2+d^2)}\,(d,\,0,\,x) = \frac{\mu_0 I}{2\pi d}\left(\frac{1}{1+\tau^2},\,0,\,\frac{\tau}{1+\tau^2}\right), \tag{10}$$

with $\tau = x/d$, which is (3). The opposite current direction changes the sign of K_p. For a straight section of finite half-length ℓ on each side of the CPA, the field at the CPA is multiplied by $\ell/\sqrt{\ell^2 + d^2}$: at $\ell = 5d$ the infinite-line model is accurate to 2%.

A.2 Expansion of the Total Field and Its Phase

A scalar magnetometer, or the norm of a vector magnetometer, measures $|\mathbf{T} + \mathbf{B}_s|$, where $\mathbf{T}$ is the geomagnetic field. With $u = 2\,\mathbf{B}_s \cdot \hat{\mathbf{T}}/|\mathbf{T}| + |\mathbf{B}_s|^2/|\mathbf{T}|^2$ we have $|\mathbf{T} + \mathbf{B}_s| = |\mathbf{T}|\sqrt{1+u} = |\mathbf{T}|(1 + u/2 - u^2/8 + O(u^3))$, so, keeping terms up to second order in $|\mathbf{B}_s|/|\mathbf{T}|$,

$$S = |\mathbf{T} + \mathbf{B}_s| - |\mathbf{T}| = \mathbf{B}_s \cdot \hat{\mathbf{T}} + \frac{|\mathbf{B}_s|^2 - (\mathbf{B}_s \cdot \hat{\mathbf{T}})^2}{2|\mathbf{T}|} + O\!\left(\frac{|\mathbf{B}_s|^3}{|\mathbf{T}|^2}\right), \qquad \hat{\mathbf{T}} = \frac{\mathbf{T}}{|\mathbf{T}|}. \tag{11}$$

The second-order terms equal $|\mathbf{B}_{s\perp}|^2/(2|\mathbf{T}|)$, where $\mathbf{B}_{s\perp}$ is the part of $\mathbf{B}_s$ perpendicular to $\mathbf{T}$; they are at most $K_p^2/(2|\mathbf{T}|) = 100^2/(2 \times 47\,994) = 0.10$ nT at $d = 2$ m, i.e. 0.1% of the anomaly. Only the first term is kept in the analysis; all simulations use the exact norm.

The horizontal part of $\hat{\mathbf{T}}$ has magnitude $\cos\delta$ and points to magnetic north, and $\hat{x}$ makes the angle ψ with magnetic north, so $\hat{\mathbf{T}} \cdot \hat{x} = \cos\delta\cos\psi$ and $\hat{\mathbf{T}} \cdot \hat{z} = \sin\delta$ ($\hat{\mathbf{T}} \cdot \hat{y}$ is irrelevant because $B_y = 0$). With (10), the first-order term is

$$\mathbf{B}_s \cdot \hat{\mathbf{T}} = K_p\left(\frac{\cos\delta\cos\psi}{1+\tau^2} + \frac{\sin\delta\,\tau}{1+\tau^2}\right) = \sqrt{\frac{\pi}{2}}\,K_p\big(\cos\delta\cos\psi\,\Phi_1 + \sin\delta\,\Phi_2\big), \tag{12}$$

which is (5), since $1/(1+\tau^2) = \sqrt{\pi/2}\,\Phi_1$ and $\tau/(1+\tau^2) = \sqrt{\pi/2}\,\Phi_2$. Its amplitude and phase in the sense of (6) are

$$a = \sqrt{A_1^2 + A_2^2} = \sqrt{\frac{\pi}{2}}\,K_p\sqrt{\cos^2\delta\cos^2\psi + \sin^2\delta}, \qquad \tan\beta = \frac{A_2}{A_1} = \frac{\tan\delta}{\cos\psi}, \tag{13}$$

with β in the quadrant of (A_1, A_2). Because $|\cos\psi| \le 1$, $|\tan\beta| \ge \tan\delta$, hence $\delta \le |\beta| \le 180° - \delta$. At Brest $\tan\delta = 43\,023/21\,271 = 2.023$ ($\delta = 63.7°$); for $\psi = 45°$, $\tan\beta = 2.023/0.707 = 2.86$, so $\beta = 70.7°$ and $a = 0.95\,\sqrt{\pi/2}\,K_p$.

A.3 Standard Integrals and Orthonormality

The integrals used throughout are

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}\tau}{(1+\tau^2)^2} = \frac{\pi}{2}, \qquad \int_{-\infty}^{\infty} \frac{\mathrm{d}\tau}{(1+\tau^2)^3} = \frac{3\pi}{8}, \qquad \int_{-\infty}^{\infty} \frac{\tau^2\,\mathrm{d}\tau}{(1+\tau^2)^4} = \frac{\pi}{16}. \tag{14}$$

Thus $\int \Phi_1^2 = (2/\pi)(\pi/2) = 1$, and since $\tau^2/(1+\tau^2)^2 = 1/(1+\tau^2) - 1/(1+\tau^2)^2$, $\int \Phi_2^2 = (2/\pi)(\pi - \pi/2) = 1$. The product $\Phi_1\Phi_2$ is odd, so $\int \Phi_1\Phi_2 = 0$: the pair (4) is orthonormal over the real line. The last two integrals of (14) give the unit energy of the gradiometer basis (9).

A.4 The Analytic Template: Hilbert Pair, Spectrum and Poles

With the Fourier convention $\hat{f}(\omega) = \int f(\tau)\, e^{-i\omega\tau}\,\mathrm{d}\tau$, the transform of $1/(1+\tau^2)$ is $\pi e^{-|\omega|}$, so

$$\hat{\Phi}_1(\omega) = \sqrt{2\pi}\, e^{-|\omega|}, \qquad \hat{\Phi}_2(\omega) = -i\,\mathrm{sgn}(\omega)\,\hat{\Phi}_1(\omega), \tag{15}$$

the second because the Hilbert transform multiplies a spectrum by $-i\,\mathrm{sgn}\,\omega$, and the inverse transform of $-i\pi\,\mathrm{sgn}(\omega)\, e^{-|\omega|}$ is $\tau/(1+\tau^2)$; hence $\Phi_2 = \mathcal{H}\{\Phi_1\}$. Two consequences follow.

First, $|\hat{\Phi}_2| = |\hat{\Phi}_1|$, so by the Wiener–Khinchin theorem Φ_1 and Φ_2 have the same autocorrelation function. Equivalently, the analytic template $\Psi = \Phi_1 + i\Phi_2$ of (7) has the one-sided spectrum

$$\hat{\Psi}(\omega) = (1 + \mathrm{sgn}\,\omega)\hat{\Phi}_1(\omega) = \begin{cases} 2\sqrt{2\pi}\, e^{-\omega}, & \omega > 0, \\ 0, & \omega < 0, \end{cases} \qquad \int \Psi(\tau)\,\Psi^*(\tau - t)\,\mathrm{d}\tau = \frac{4}{2 - it}. \tag{16}$$

Correlating a channel with Ψ is therefore a quadrature receiver, and the modulus of (16), $4/\sqrt{4+t^2}$, carries no dependence on the phase β of (8): this is the phase invariance exploited in Section **??**, and what record truncation destroys.

Second, $\Psi(\tau) = \sqrt{2/\pi}/(1 - i\tau)$ continues analytically off the real axis as a rational function with the single pole $\tau = -i$. A shifted template $\Psi(\tau - t)$ has its pole at $t - i$, and two rational functions that agree on an interval agree everywhere, so their poles must coincide: for real t this forces $t = 0$. This is the argument used in Appendix **??** to show that the subspace coherence of a window with $\mathrm{span}\{\Phi_1, \Phi_2\}$ reaches one at the CPA and nowhere else, and that a wrong template scale λ moves the pole to $t - i\lambda$ and so prevents it.